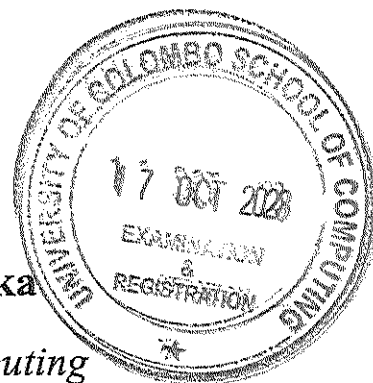




University of Colombo, Sri Lanka

University of Colombo School of Computing

BACHELOR OF SCIENCE IN COMPUTER SCIENCE
BACHELOR OF SCIENCE HONOURS IN COMPUTER SCIENCE
BACHELOR OF SCIENCE HONOURS IN SOFTWARE ENGINEERING

Third Year Examination - Semester I - **UCSC AY20** [held in October 2023]

SCS3206 — Graph Theory

(Two (2) Hours)

Answer ALL questions

Number of Pages = 17

Number of Questions = 4

To be completed by the candidate

Index Number:

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97

Important Instructions to candidates:

- I. The medium of instructions and questions is in English.
- II. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- III. MCQ should be marked on the **answer sheet** and **structured question** on the **space provided in the question paper**.
- IV. Write your index number **CLEARLY** on each and every page of this Question paper.
- V. This paper consists of 4 questions in 17 pages (including the Cover Page).
- VI. Answer **ALL** questions.
- VII. Programmable Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are not allowed.
- VIII. Non-Programmable calculators are allowed.
- IX. Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate.

To be completed by the examiners

1	
2	
3	
4	
Total	

Index Number -

The following MCQ questions in Question 01 and Question 02 have 05 (five) choices with one or more correct answers.

Mark the correct choices on the question paper first and then transfer them to the given answer sheet.

Question 01

The mark given for a question will vary from 0 (All the incorrect choices are marked & no correct choices are marked) to +1 (All the correct choices are marked & no incorrect choices are marked).

1. Which of the following statement(s) is/are **True** regarding a **simple graph**?

- I. The total degree of a simple graph is even.
- II. There are odd number of vertices with even degrees.
- III. When considering the adjacency matrix, the degree of a vertex is equal to the number of 1's in the corresponding row of the matrix.

- (i) I only
- (ii) II and III only
- (iii) I and II only
- (iv) I and III only
- (v) All I, II, and III

2. Which of the following statement(s) is/are **True** regarding **graph data structures**?

- (i) Arrays are efficient to be used to check if two vertices u and v in a graph are connected.
- (ii) Arrays are efficient to be used when retrieving all the neighbors of a vertex u .
- (iii) Lists (Linked Lists) are efficient to be used to check if two vertices u and v in a graph are connected.
- (iv) It is efficient to use an array when accessing its n -th non-zero element.
- (v) None of the above are correct.

3. A **graph traversal** is a method to:

- I. Split the graph in half.
- II. Explore the graph, one vertex at a time, using knowledge about the neighbors of explored vertices.
- III. Obtain the list of accessible vertices from an initial one.

- (i) II only
- (ii) III only
- (iii) I and II only
- (iv) II and III only
- (v) All I, II, and III

Index Number -

4. How many regions (i.e., A space that is divided by the edges and vertices of a planar graph) can be observed in a connected planar graph with 5 vertices and 9 edges?
- (i) 5
 - (ii) 6
 - (iii) 8
 - (iv) 9
 - (v) It could be different according to different planar representations.
5. Which of the following statement(s) is/are **True** regarding **regular graphs**?
- I. It is possible to have a 3-regular graph with 12 vertices.
 - II. Every complete graph K_n is an $(n-1)$ regular graph.
 - III. The number of edges in a cubic graph with 10 vertices is 15.
- (i) I only
 - (ii) II only
 - (iii) I and II only
 - (iv) II and III only
 - (v) All I, II, and III
6. Which traversal technique(s) visit the nodes in **ascending order** in a **binary search tree**?
- (i) Inorder
 - (ii) Preorder
 - (iii) Postorder
 - (iv) Level-order
 - (v) None of the above
7. Which method(s) is/are used to calculate the **number of spanning trees**?
- (i) Hall's theorem
 - (ii) Euler's theorem
 - (iii) Prims Algorithm
 - (iv) Kirchhoff's theorem
 - (v) Handshaking theorem
8. In **Kruskal's algorithm**, which **data structure** is commonly used to efficiently determine if adding an edge will create a cycle?
- (i) Queue
 - (ii) Stack
 - (iii) Hash Table
 - (iv) Priority Queue
 - (v) Union-Find Data Structure

Index Number -

9. Which algorithm(s) is/are used to solve the **maximum flow problem**?

- (i) Prim's algorithm
- (ii) Dinic's algorithm
- (iii) Kruskal's algorithm
- (iv) Dijkstra's algorithm
- (v) Bellman-Ford algorithm

10. What is the **maximum flow** in a flow network?

- (i) The minimum cut of a network
- (ii) The total number of edges in the network.
- (iii) The total capacity of all edges in the network.
- (iv) The capacity of the edge with the maximum flow.
- (v) The minimum capacity of any edge in the network.

Question 02

The mark given for a question will vary from 0 (All the incorrect choices are marked & no correct choices are marked) to +2 (All the correct choices are marked & no incorrect choices are marked).

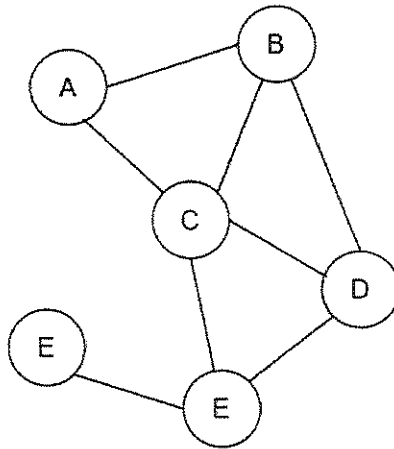
11. Which of the following proposition(s) is/are **True** regarding the application of **Depth-First Search (DFS)** from a vertex **v1** of an unweighted graph?

- (i) A spanning tree, covering the accessible vertices from v1 can be obtained.
- (ii) The shortest paths from v1 to the accessible vertices of the graph can be obtained.
- (iii) We follow a trail, as long as there are unexplored vertices to jump to.
- (iv) Each edge of the graph is visited exactly one time.
- (v) None of the above are correct.

12. Which of the following proposition(s) is/are **True** regarding the application of **Breadth-First Search (BFS)** from a vertex **v1** of an unweighted graph?

- (i) A spanning tree, covering the accessible vertices from v1 can be obtained.
- (ii) The shortest paths from v1 to the accessible vertices of the graph can be obtained.
- (iii) We follow a trail, as long as there are unexplored vertices to jump to.
- (iv) Each edge of the graph is visited exactly one time.
- (v) None of the above are correct.

13 Which of the following statement(s) is/are **False** regarding the **coloring of the following graph**?



- (i) The edge-chromatic number of the above graph is 4.
- (ii) The edge-chromatic number of the above graph is 5.
- (iii) The chromatic number of the above graph is 3.
- (iv) The chromatic number of the above graph is 4.
- (v) The total chromatic number of the above graph is 7.

14 Which of the following statement(s) is/are **False** regarding **simple graphs**?

- (i) The maximum possible degree of a simple graph is $n-2$, where n is the number of vertices of the graph.
- (ii) The maximum possible degree of a simple graph is $n-1$, where n is the number of vertices of the graph.
- (iii) The maximum possible number of edges in a simple graph is $n(n-1)$ where n is the number of vertices of the graph.
- (iv) A simple graph may not have loops.
- (v) A simple graph may not have multiple edges connecting the same vertex pair.

15 Which of the following statement(s) is/are **True** regarding **Dijkstra's** and **A* algorithms**?

- I. Both algorithms may return $-\infty$ as a distance if there's a negative cycle or a negative self-loop in the graph.
 - II. Dijkstra's algorithm is generally more efficient than the A* algorithm.
 - III. Both algorithms will return the shortest path from a single vertex to every other vertex in a graph.
- (i) I only
 - (ii) III only
 - (iii) II and III only
 - (iv) I and II only
 - (v) I and III only

Index Number -

16 Which of the following statement(s) is/are **True** regarding the **Bellman-Ford** and **Floyd-Warshall algorithms**?

- I. Both are single source shortest path (SSSP) algorithms.
- II. Both algorithms can detect negative cycles in a graph.
- III. The Floyd-Warshall algorithm is more computationally efficient than the Bellman-Ford algorithm.

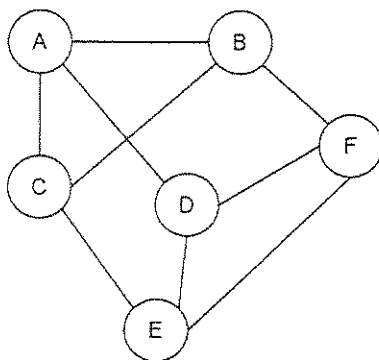
- (i) I only
- (ii) II only
- (iii) I and II only
- (iv) II and III only
- (v) I and III only

17 Which of the following statement(s) is/are **True** regarding the **Dijkstra's** and **Bellman-Ford algorithms**?

- I. Both algorithms will find the optimal path from a given node to another node even if a graph has negative edge weights.
- II. Dijkstra's algorithm is more computationally efficient than the Bellman-Ford algorithm.
- III. Both can be considered as greedy algorithms.

- (i) II only
- (ii) III only
- (iii) I and II only
- (iv) II and III only
- (v) All the statements are False.

18 Which of the following statement(s) is/are **True** regarding the following graph?

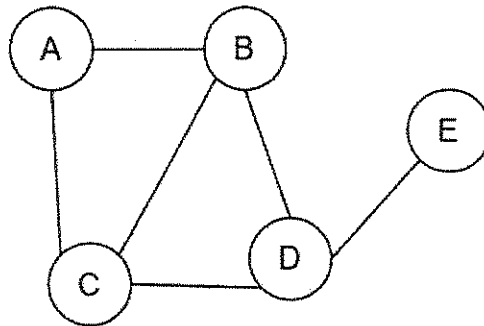


- I. The diameter of the graph is 2.
- II. The radius of the graph is 1.
- III. The eccentricity of vertex A is 3.
- IV. B is a central vertex of the graph.

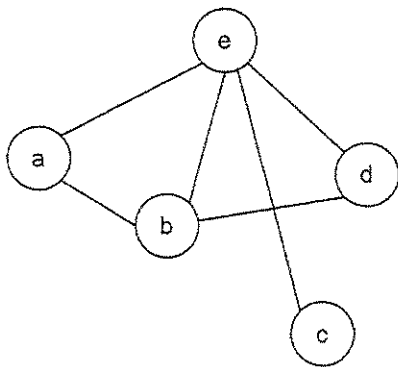
- (i) I only
- (ii) I and II only
- (iii) I and IV only
- (iv) I, II and III only
- (v) I, II and IV only

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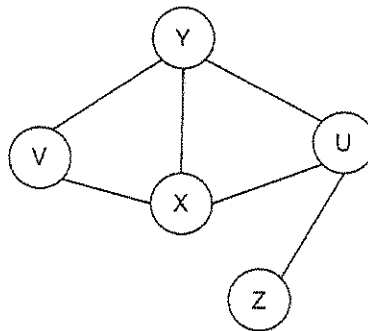
19 Which of the following graph(s) is/are **isomorphic** with the given graph?



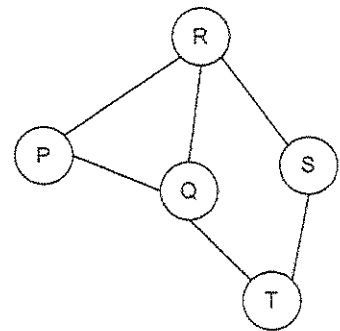
(i)



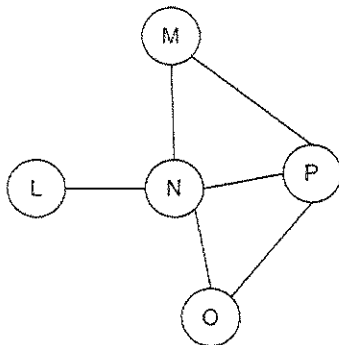
(ii)



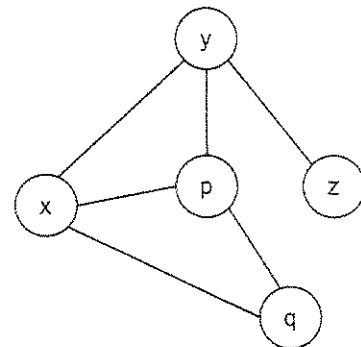
(iii)



(iv)



(v)



20 Which of the following statement(s) is/are **True**?

- I. Multiple graphs can be created with the same degree sequence.
- II. If a graph is disconnected, it may have more than one connected component.
- III. A trail of a graph may contain repeated vertices but not repeated edges.

- (i) I only
- (ii) II only
- (iii) I and II only
- (iv) II and III only
- (v) All I, II, and III

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- 21 Which of the following degree sequence(s) represent a tree with five (05) vertices?
- (i) 1, 1, 1, 1, 1
 - (ii) 1, 1, 1, 2, 2
 - (iii) 1, 1, 2, 2, 2
 - (iv) 1, 1, 2, 2, 3
 - (v) 1, 1, 2, 3, 4
- 22 How many vertices are there in a forest consisting of k trees if the forest comprises n edges?
- (i) nk
 - (ii) $n - k$
 - (iii) $n + k$
 - (iv) $kn - k$
 - (v) $kn + k$
- 23 What is the total number of edges in a tree with the degree sequence: 1, 1, 1, 1, 2, 3, 3?
- (i) 6
 - (ii) 7
 - (iii) 8
 - (iv) 12
 - (v) 14
- 24 Which of the following statement(s) is/ are True,
- I. If we add any edge to a tree, it will create a cycle.
 - II. The leaf is the only vertex having a degree of 1 in a tree.
 - III. The summation of the degree of a tree with n vertices will be $2n - 2$.
- (i) I only
 - (ii) I and II only
 - (iii) I and III only
 - (iv) II and III only
 - (v) All I, II, and III
- 25 Which of the following statement(s) is/are True regarding Trees?
- I. A tree can contain isolated vertex.
 - II. A tree with n edges can have $n+1$ vertices.
 - III. Any two vertices will have only one path in a tree.
- (i) I only
 - (ii) I and II only
 - (iii) I and III only
 - (iv) II and III only
 - (v) All I, II, and III

26 Which of the following statement(s) is/are **True** regarding **Binary Trees**.

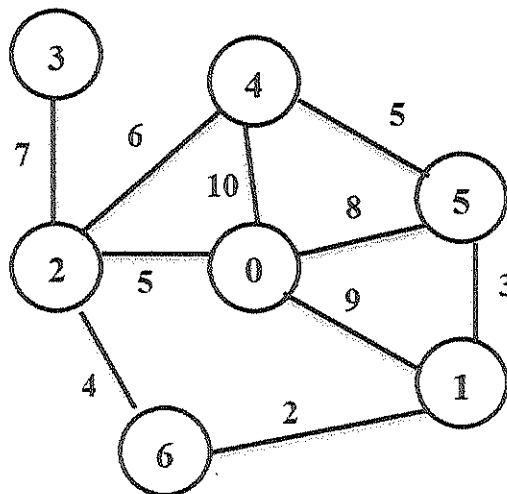
- I. A perfect binary tree of height h has $2^{h+1} - 1$ vertices.
- II. A binary tree has at most 1 child for each internal node.
- III. In a binary tree, the node that has no children is called a leaf node.

- (i) I only
- (ii) I and II only
- (iii) I and III only
- (iv) II and III only
- (v) All I, II, and III

27 If we have a tree with 5 vertices labeled as A, B, C, D, and E, how many **distinct labeled trees** can be generated using the same set of labels (A, B, C, D, E)?

- (i) 25
- (ii) 32
- (iii) 64
- (iv) 125
- (v) 625

28 What is the **summation of the weights** in the **minimum spanning tree** of the graph below,



- (i) 21
- (ii) 26
- (iii) 30
- (iv) 39
- (v) 59

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29 Which of the following statement(s) is/are **True** regarding **Graph Isomorphism**,

- I. Graphs and trees can have more than two centers.
- II. Two graphs can be isomorphic if they are structurally similar.
- III. When serializing a tree into a unique string, the labels of the previous subtree will be sorted lexicographically and wrapped in brackets.

- (i) I only
- (ii) I and II only
- (iii) I and III only
- (iv) II and III only
- (v) All I, II, and III

30 Which of the following statement(s) is/are **True** regarding **Flow Networks**,

- I. Residual edges have a capacity of zero.
- II. Flow can be greater than or equal to the capacity of a given edge.
- III. Capacity is the maximum amount of flow that can pass through the edge.

- (i) I only
- (ii) I and II only
- (iii) I and III only
- (iv) II and III only
- (v) All I, II, and III

Cont.

Question 03

1. State two (02) applications of trees.

[2 marks]

2. The following questions are based on the adjacency matrix given below which represents graph G1.

$$G1 = \begin{bmatrix} 0 & 2 & 5 & 2 & 3 & 0 \\ 2 & 0 & 0 & 1 & 0 & 0 \\ 5 & 0 & 0 & 1 & 6 & 0 \\ 2 & 1 & 1 & 0 & 4 & 8 \\ 3 & 0 & 6 & 4 & 0 & 0 \\ 0 & 0 & 0 & 8 & 0 & 0 \end{bmatrix}$$

- (a) State whether the G1 is a directed or undirected graph, Justify your answer.

[2 marks]

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- (b) Compare the **adjacency list** and **adjacency matrix** representations in terms of their suitability for graphs with **different densities**.

Sparse Graph:

Dense Graph:

[3 marks]

- (c) Draw the graph G1 by connecting the edges between the vertices given below.

0

1

2

3

4

5

[3 marks]

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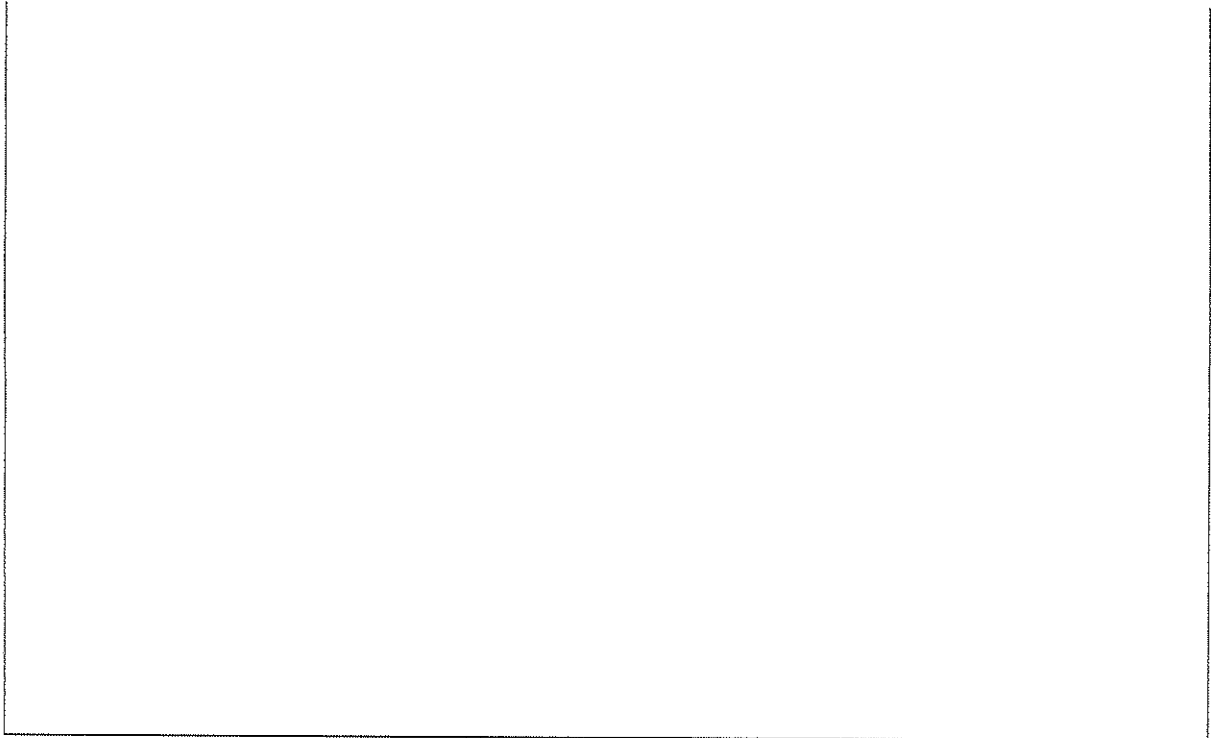
- (d) Outline the steps for computing minimum spanning tree (MST) using **Kruskal's algorithm**.

[5 marks]

- (e) Compute the **minimum spanning tree** for graph G1 using **Prim's eager version**, mentioning each step clearly.

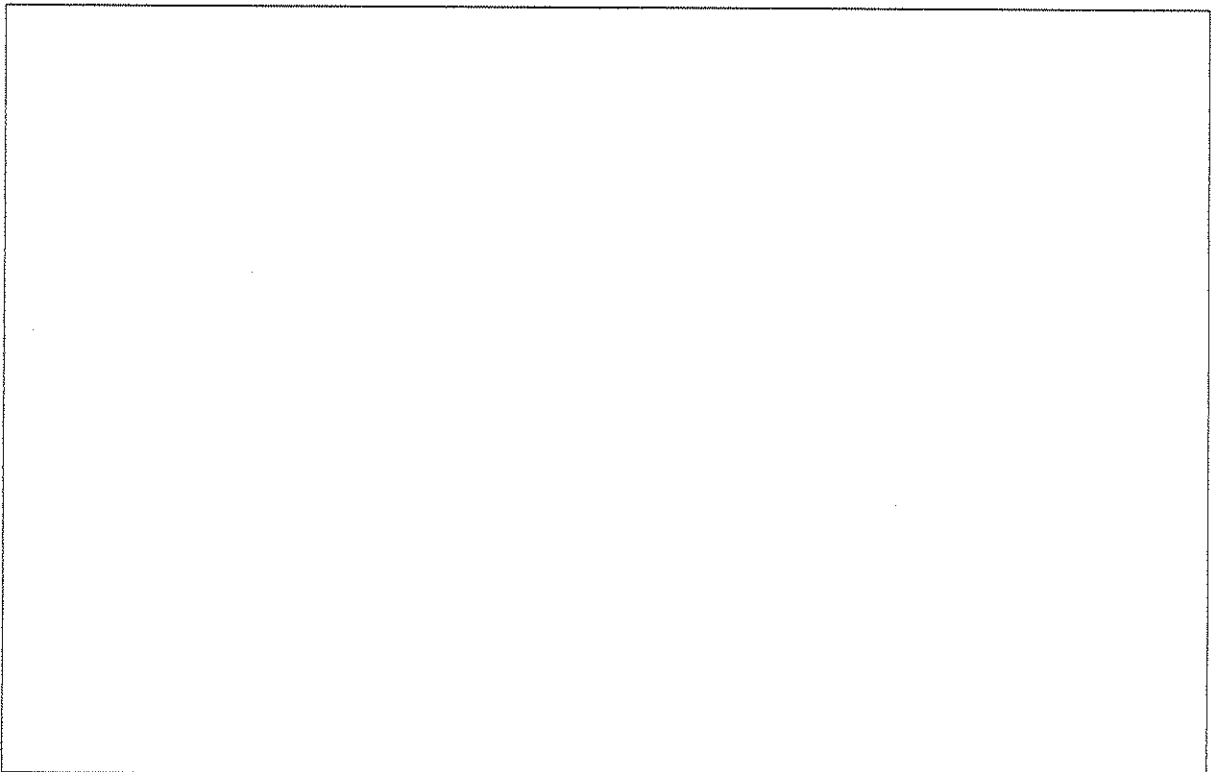
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Index Number -



[6 marks]

- (f) Compare and contrast **Prim's eager version** and **Prim's lazy version** focusing on the **edge selection process** and **data structures used**.



[4 marks]

Question 04

1.

(a) Define the following terms in the context of network flow models.

- i. Augmenting Paths
- ii. Bottleneck
- iii. Augmenting the flow

Augmenting Paths:

Bottleneck:

Augmenting the flow:

[3 marks]

(b) State the purpose of incorporating **residual edges** in network flow models.

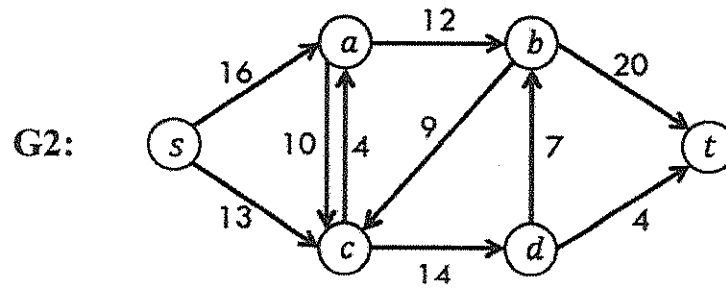
[2 marks]

(c) Compare the **Edmonds-Karp algorithm** and **Capacity Scaling algorithm** in terms of their approaches used to calculating maximum flow in a network.

[4 marks]

Index Number -

- (d) Compute the maximum flow for Graph G2 using the Edmonds-Karp algorithm. Provide clear details about the augmenting paths chosen during the process.



[6 marks]

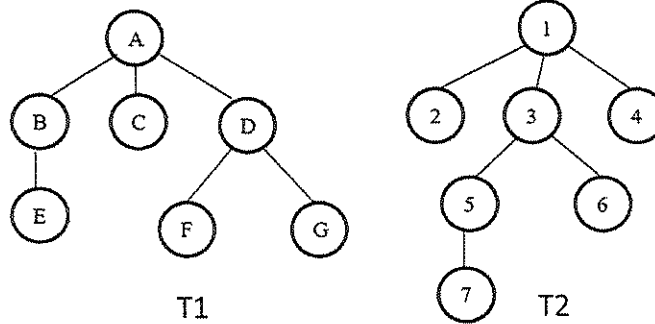
Index Number -

2.

- (a) Briefly describe a technique for determining the centers of a tree.

[2 marks]

- (b) Identify the centers of trees T1 and T2 provided below and rearrange their structures accordingly.



[4 marks]

- (c) Serialize trees T1 and T2 using AHU encoding and determine if they are isomorphic.

[4 marks]
